

Quiz Rush: A Real-Time Quiz Battle Using Discrete Logic

Problem Statement

This project presents the design and simulation of a multi-player quiz buzzer system with a two-stage priority mechanism(refer fig 1). The system detects the fastest player to press the buzzer and locks out all other players to ensure fair competition.

If the first-priority player gives an incorrect answer, the second-stage circuit is accessed only after the wrong response is detected. Once this happens, the second part of the circuit becomes enabled, allowing the remaining two players to compete, and the one who presses the buzzer first in this stage gets the opportunity to answer.

Answer validation is performed using AND logic. When the answer is correct, the AND gate output goes HIGH and is fed into an OR gate. The OR gate output drives a green LED, indicating a correct response. This HIGH signal also triggers a timer pulse that increments the score automatically using a CD4026 counter, displaying the updated score on a 7-segment display(refer fig 2).

If the answer is wrong, a NOT gate inverts the signal and a red LED glows to indicate an incorrect response. The system also includes a reset feature to synchronize each round for smooth and reliable operation.

Methodology

- **Buzzer Input Stage**
Each player has a push-button. When pressed, it sends a signal to the priority circuit.
- **Two-Stage Lockout Circuit**
The system accepts only the first buzzer press.
In Stage 1, the fastest player is selected and others are locked.
If the answer is wrong, only then the second stage is accessed, enabling the remaining players to compete.
- **Winner Indication**
An LED glows to indicate the selected player.
- **Answer Validation**
The player's signal and quizmaster's switch are given to an AND gate.
If correct, a green LED glows.
If wrong, a red LED glows.
- **Automatic Score Update**
For a correct answer, a pulse increments by (+1) the score using a CD4026 counter.
- **Score Display**
The score is shown on a 7-segment display.
- **Reset**
A reset switch clears the system for the next round.

Circuit Diagram

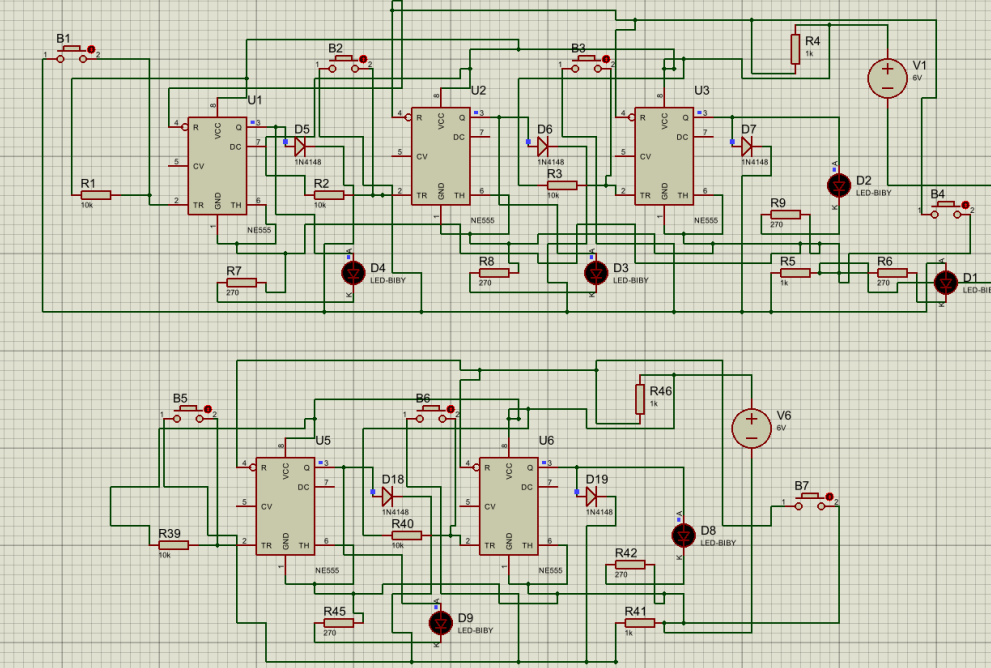


Fig 1: 1st and 2nd Priority

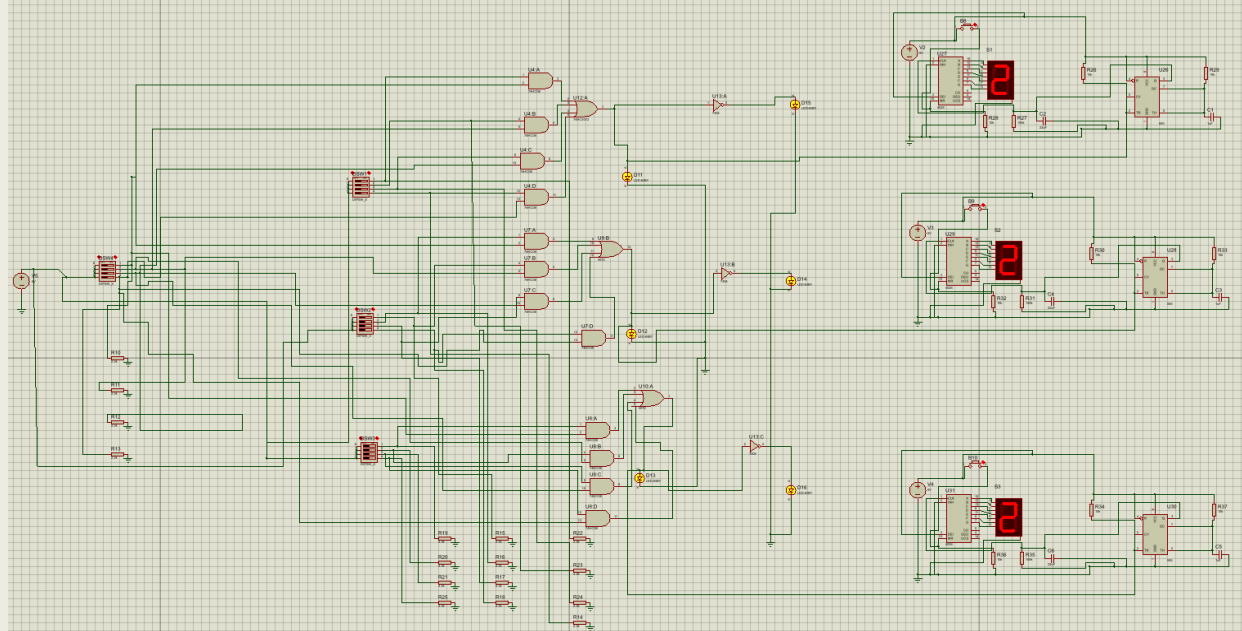


Fig 2: Answer validation and score incrementer

Simulation Results

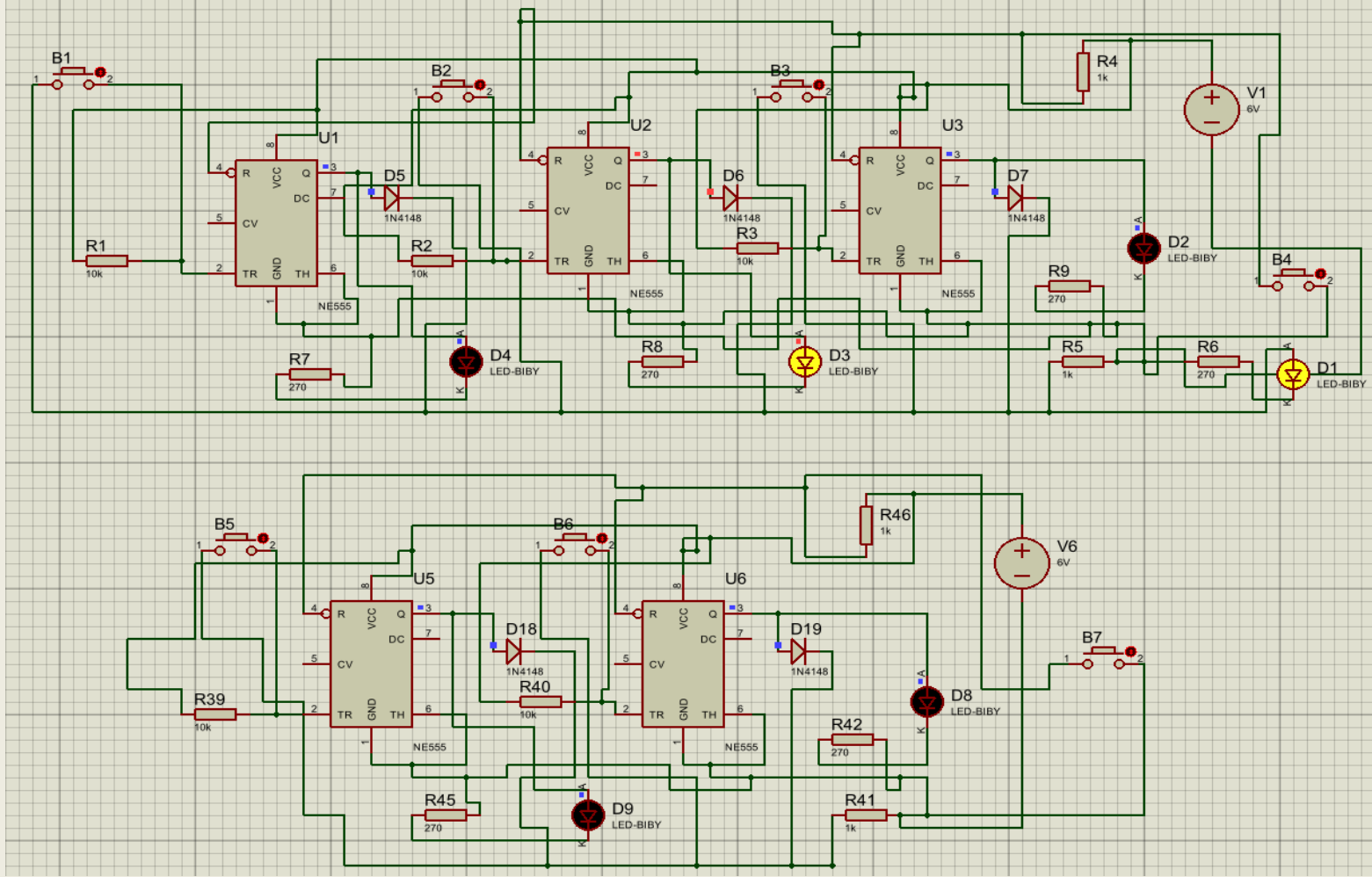


Fig 3: Output of priority circuit

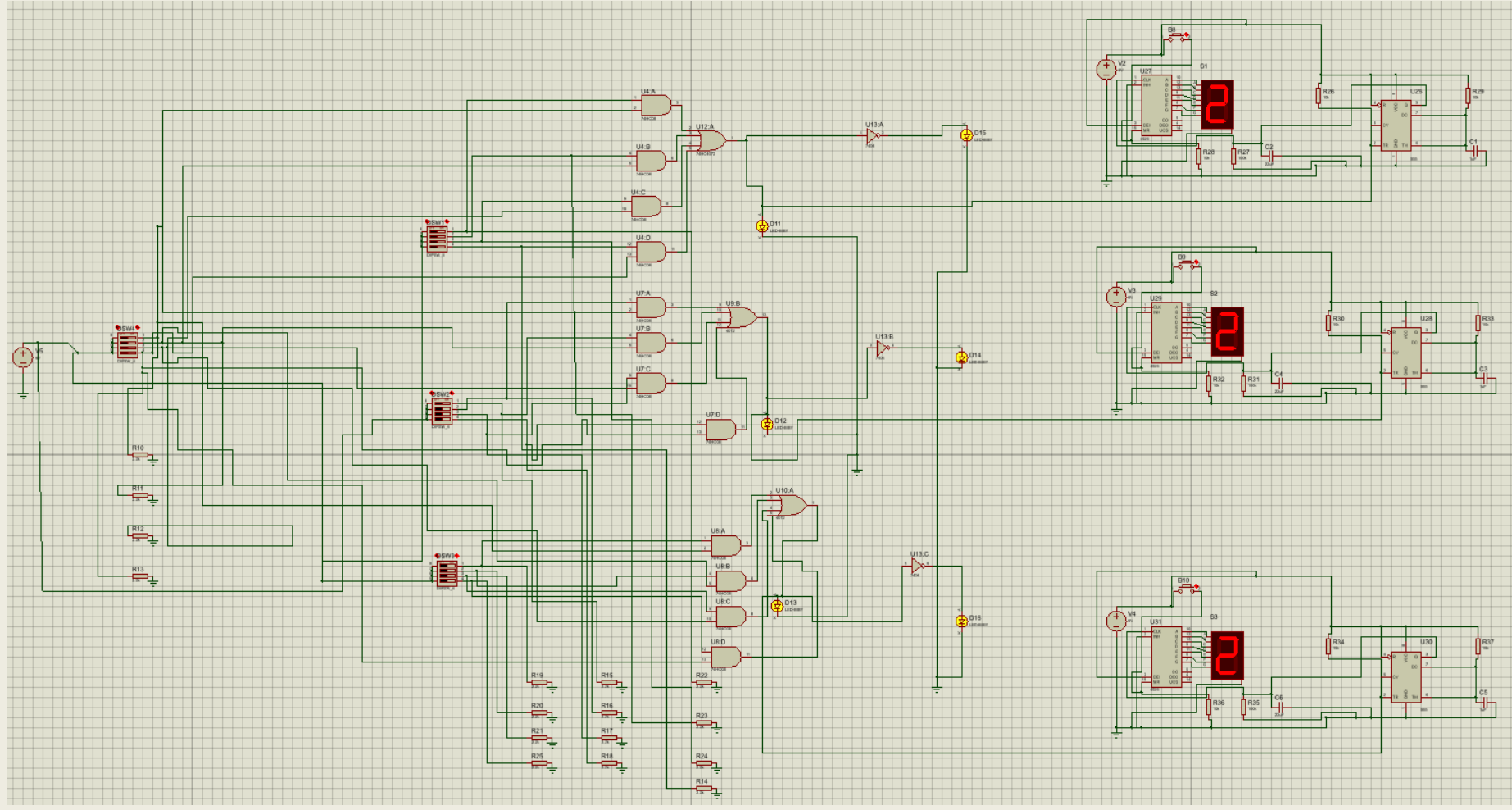


Fig 4: Output of answer validation and score incrementer

Working Model Picture

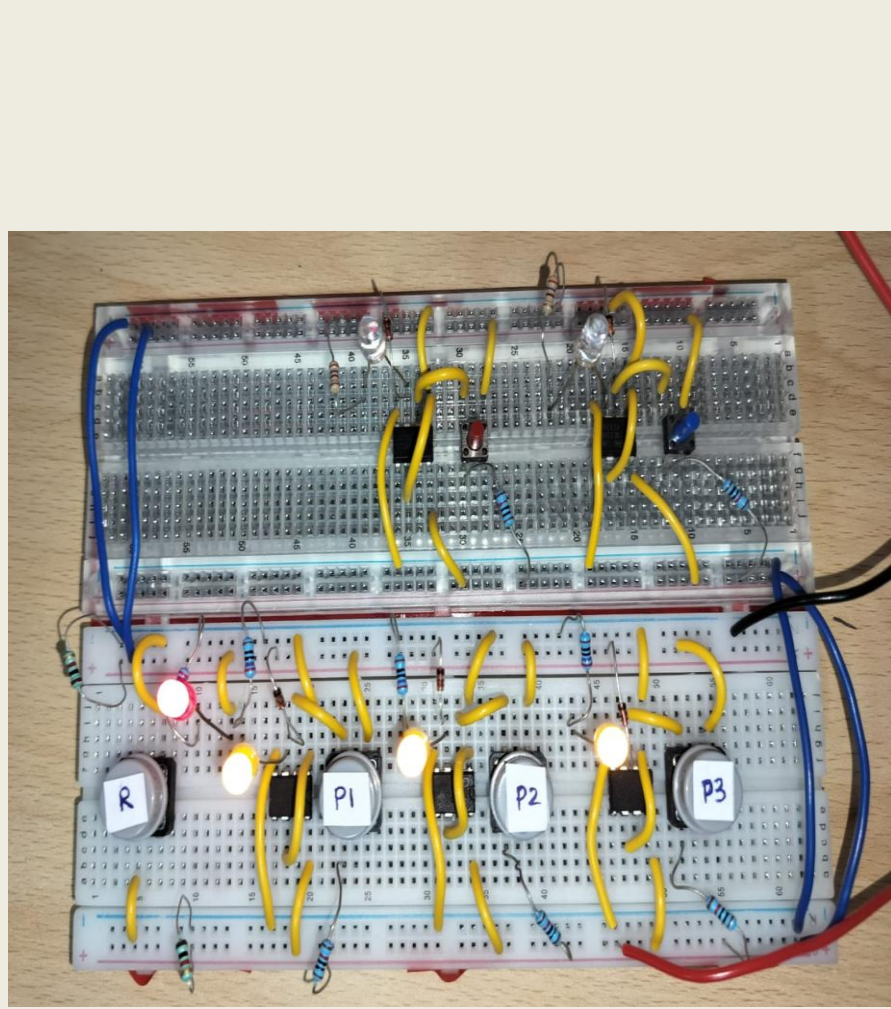


Fig 5: Working model of priority 1 and priority 2

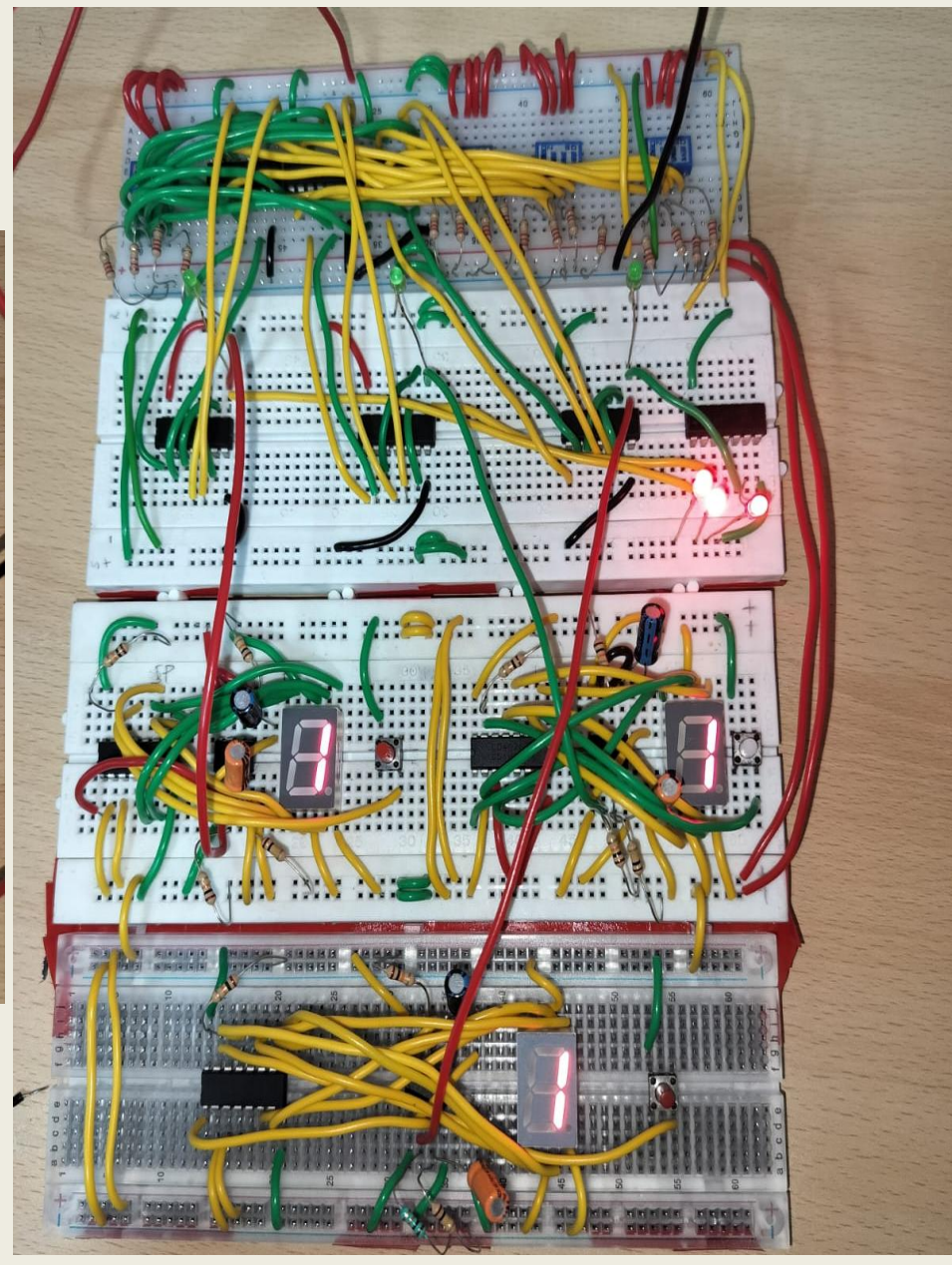


Fig 6: Hardware implementation of answer validation and score display

PCB Layout

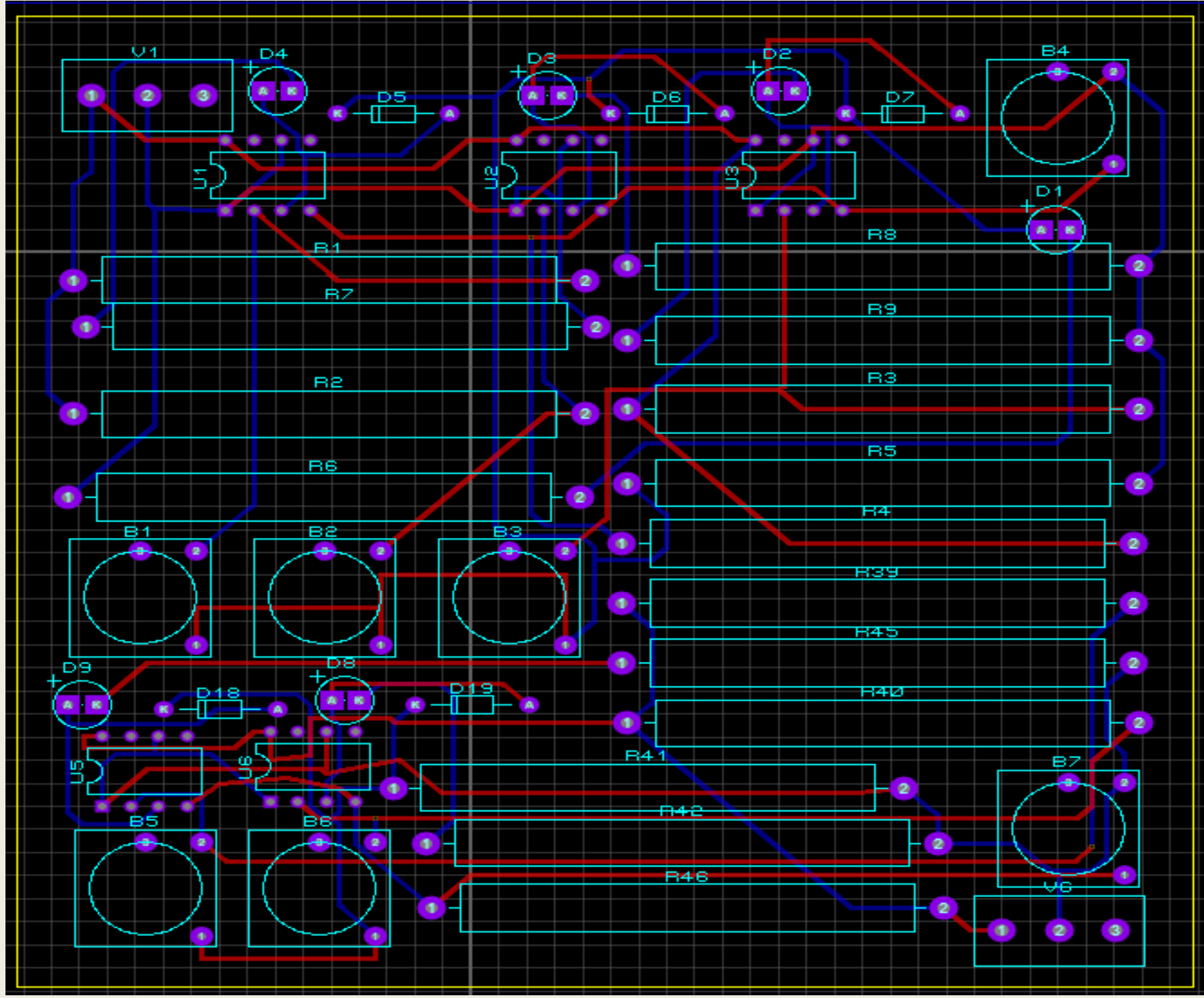


Fig 7: PCB layout for 1st and 2nd priority circuit

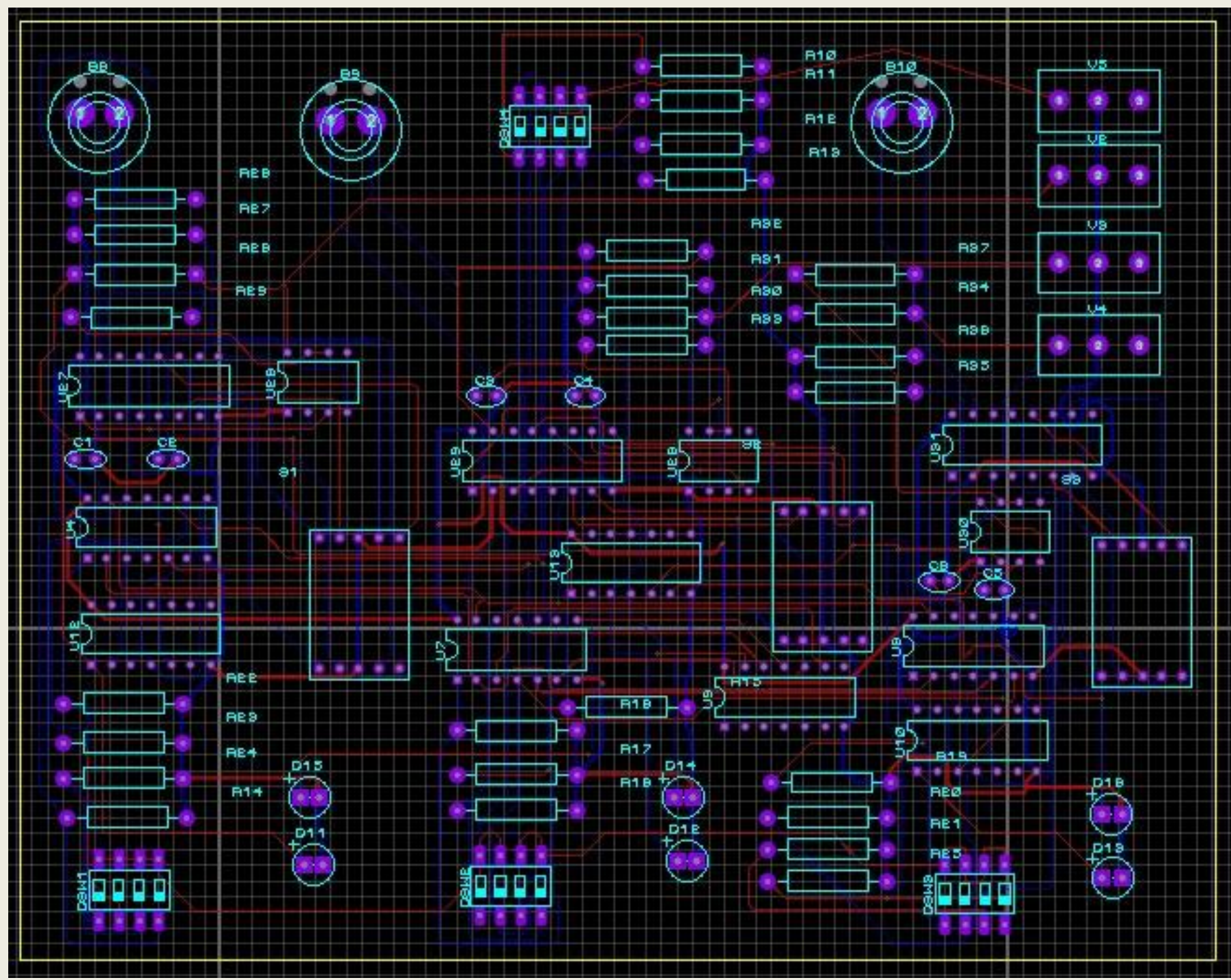


Fig 8: PCB Layout for answer validation and score incrementer circuit

Results

- The automatic scoring system was successfully designed, simulated, and implemented with complete hardware integration(refer fig 5&6).
- The buzzer input stage accurately detected the fastest player using the two-stage timer lockout circuit, ensuring proper priority selection and fair operation.
- Answer validation using AND logic generated a HIGH output for correct responses, triggering a timer pulse to increment the score automatically through the CD4026 counter.
- The updated score was displayed clearly on the 7-segment display, while incorrect responses activated the NOT gate and red LED indication.
- Winner indication LEDs and the reset mechanism functioned reliably for smooth round operation.
- The entire circuit was fabricated on a PCB (refer fig 7&8) with organized component placement, proper routing, and secure soldered connections to ensure stable signal flow.

References

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